
Exact Optimality of Communication-Privacy-Utility Tradeoffs in Distributed Mean Estimation

Berivan Isik

Stanford University

berivan.isik@stanford.edu

Wei-Ning Chen

Stanford University

wnchen@stanford.edu

Ayfer Ozgur

Stanford University

aozgur@stanford.edu

Tsachy Weissman

Stanford University

tsachy@stanford.edu

Albert No

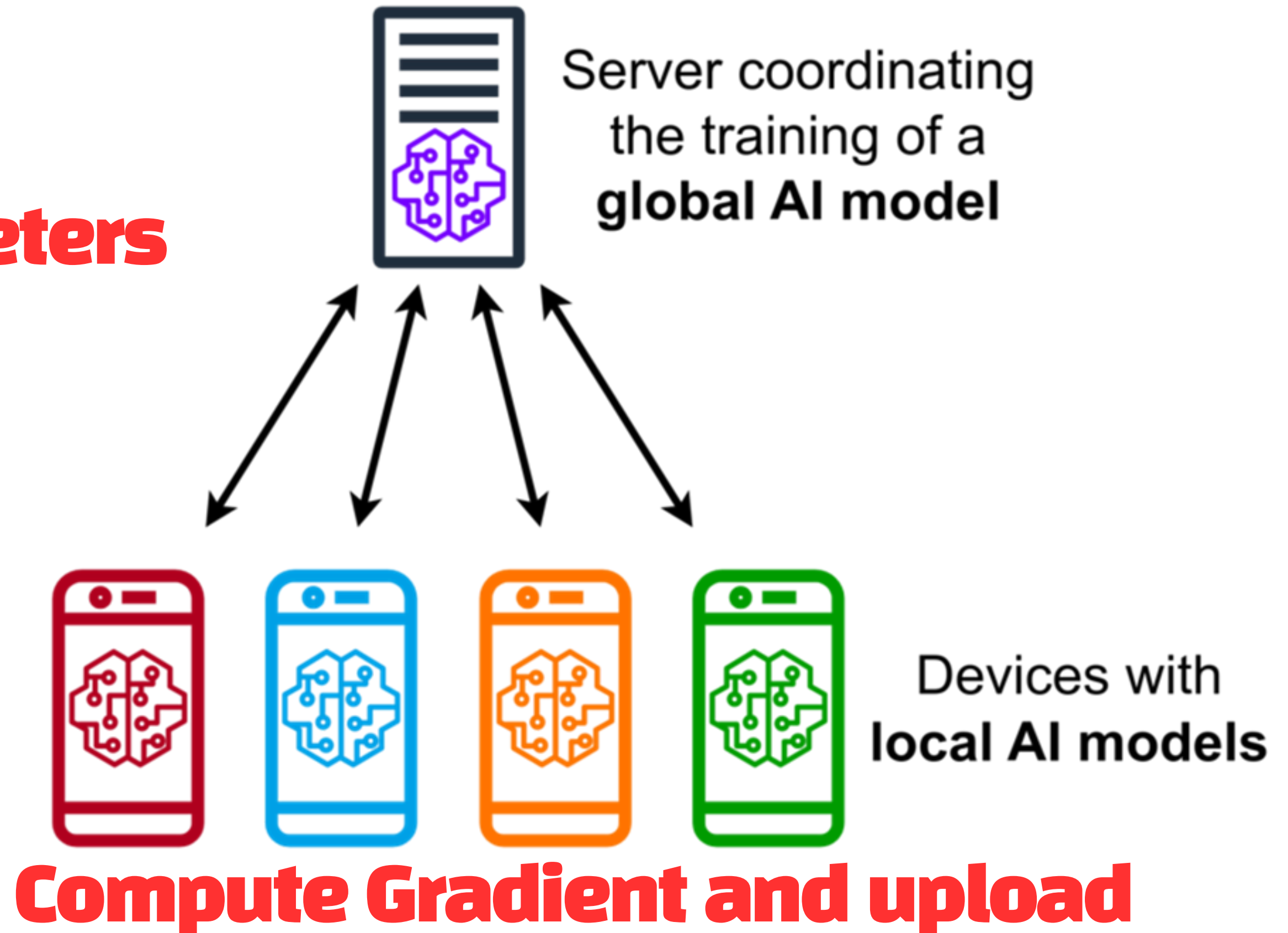
Hongik University

albertno@hongik.ac.kr

NeurIPS 2023

Federated Learning

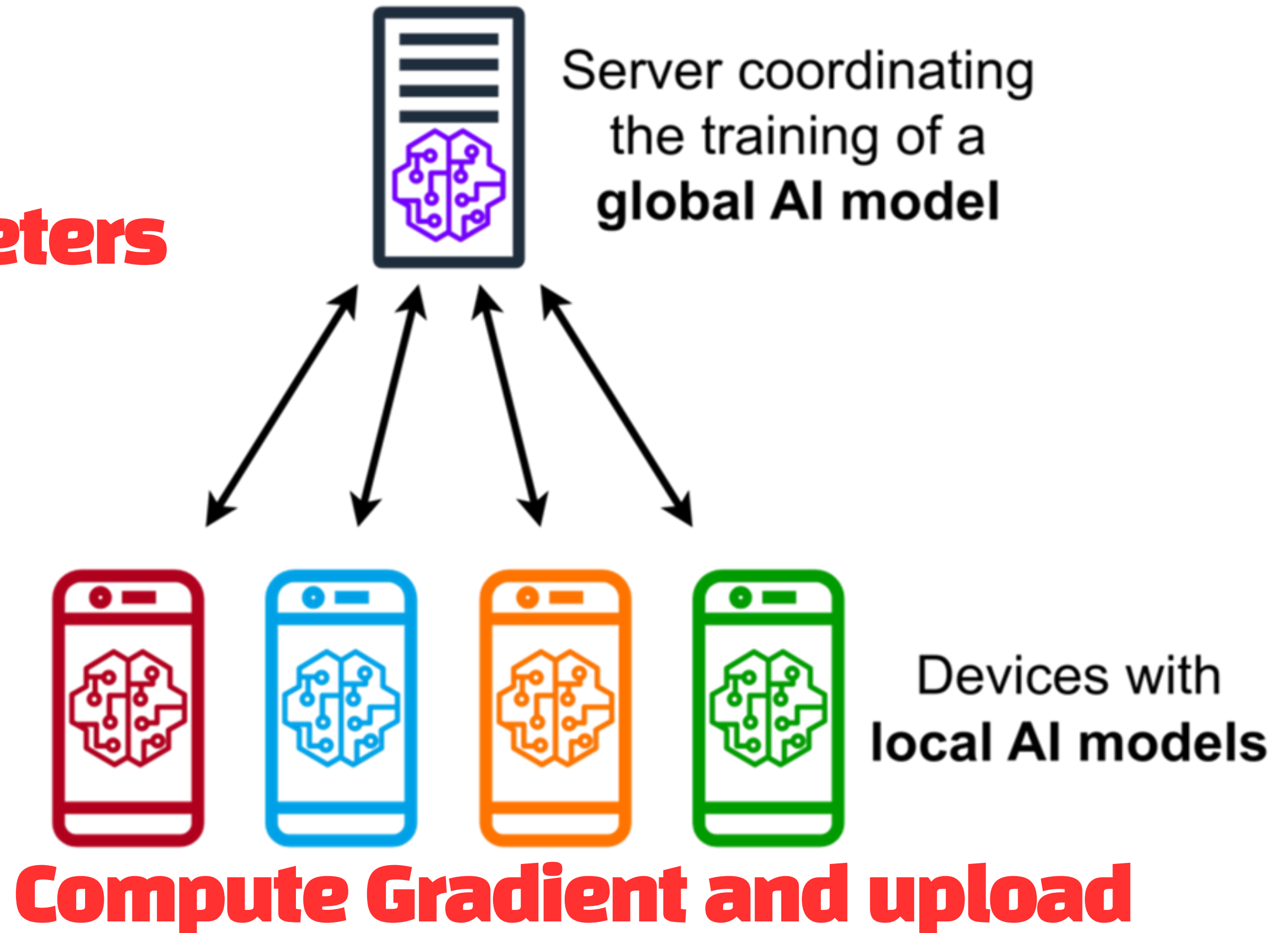
**Collect Gradient
and update parameters**



Federated Learning: Challenges

**Collect Gradient
and update parameters**

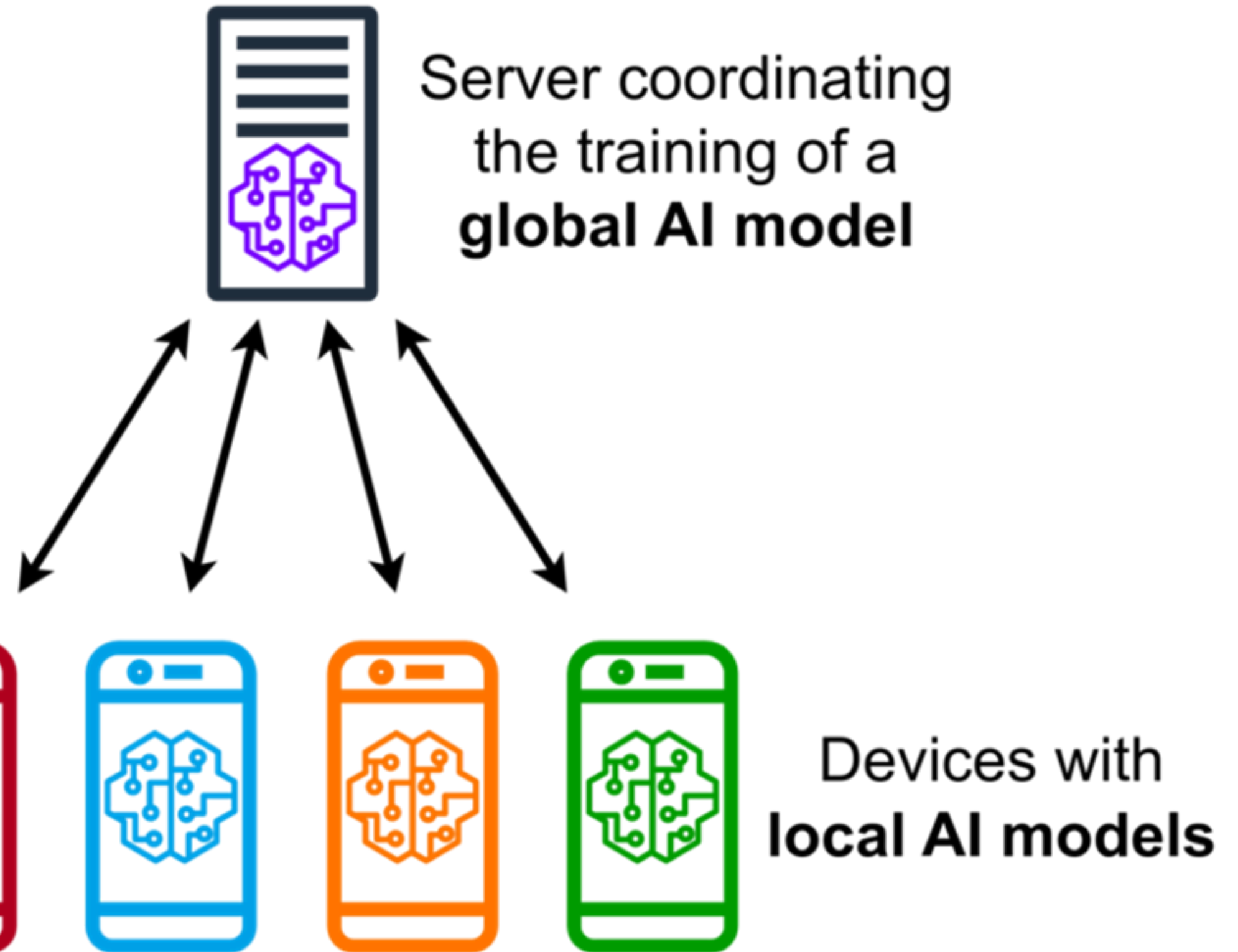
**Compression
Privacy
Utility**



Mean Estimation

**Collect vector
and compute mean**

**Compression (b bits)
Privacy (LDP)
Utility (MSE)**



Privatize & Compress vector and upload

LDP vs DP

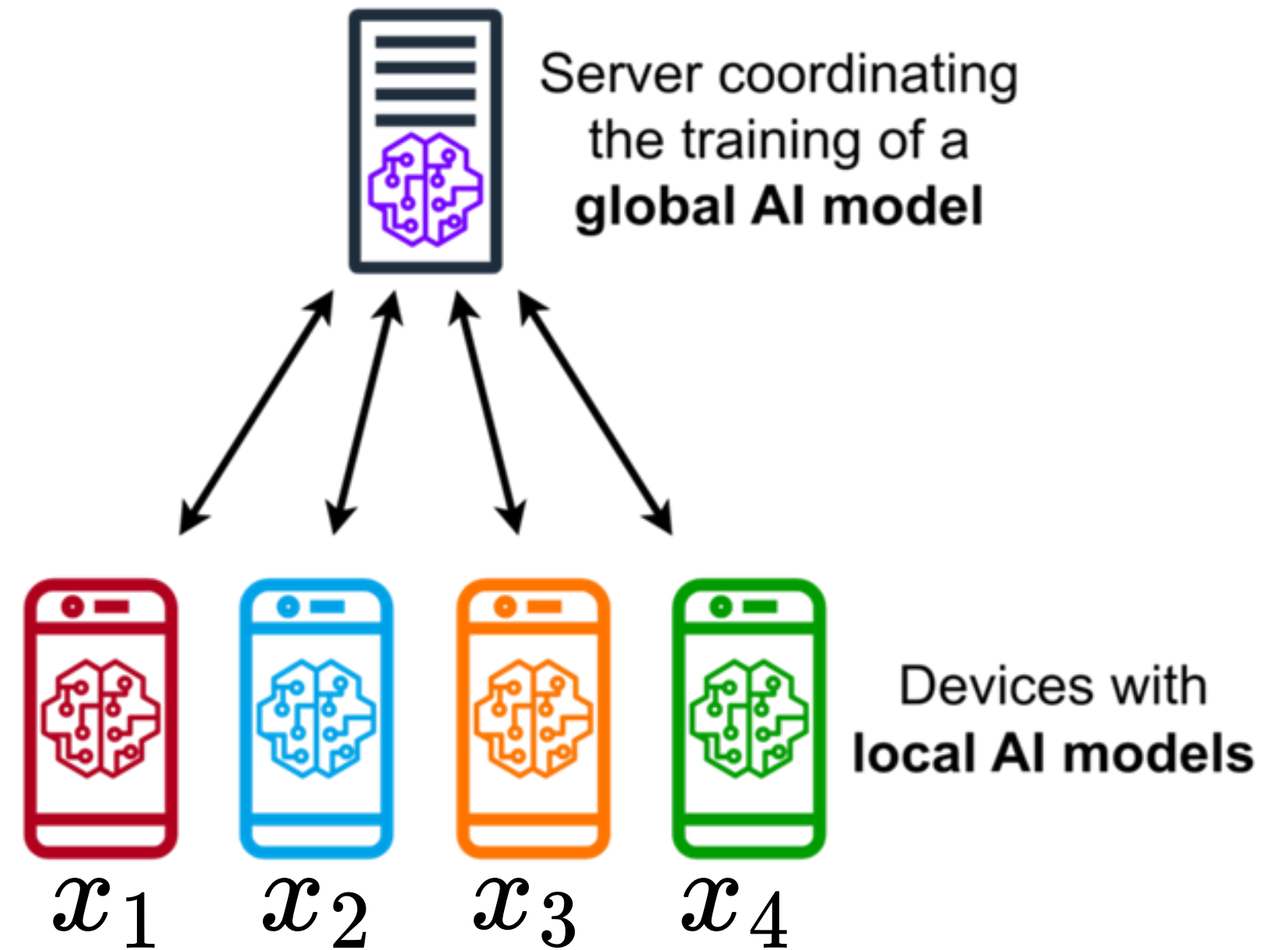
DP

- **Trusted Server**

LDP

- **Unfaithful Server**

$$m = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

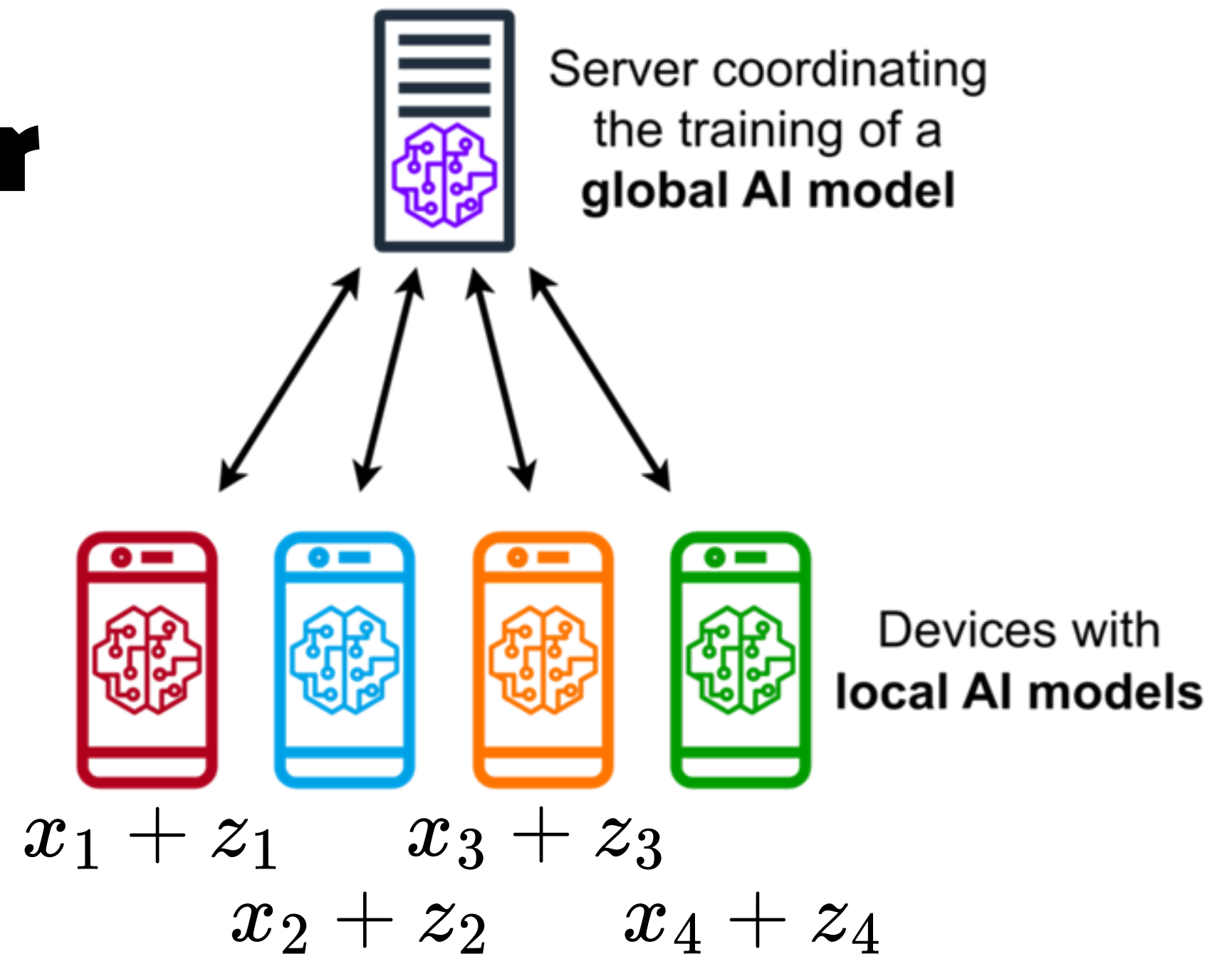


LDP: Gaussian Mechanism

$$m = \frac{x_1 + x_2 + x_3 + x_4}{4} + \frac{z_1 + z_2 + z_3 + z_4}{4}$$

Privatize individual vector

- Gaussian Mechanism is suboptimal**

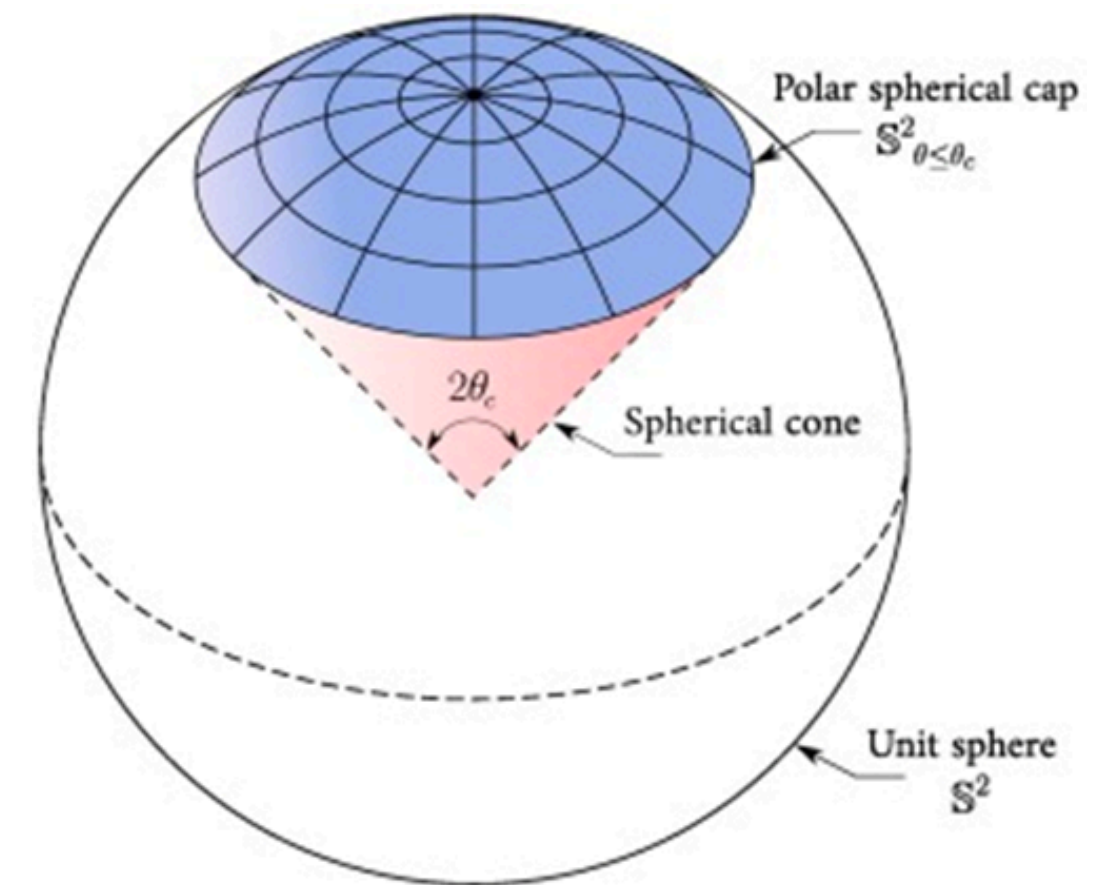


PrivUnit: Privacy vs Utility

We begin by introducing PrivUnit which was developed by Bhowmick et al. [BDFKR18]. Given an input vector $v \in \mathbb{S}^{d-1}$ and letting $W \sim \text{Uni}(\mathbb{S}^{d-1})$, $\text{PrivUnit}(p, \gamma)$ has the following distribution (up to normalization)

$$\text{PrivUnit}(p, \gamma) \sim \begin{cases} W \mid \langle W, v \rangle \geq \gamma & \text{with prob. } p \\ W \mid \langle W, v \rangle < \gamma & \text{with prob. } 1 - p \end{cases}$$

without Compression



PrivUnit is Optimal

ICML 2022 Oral

Optimal Algorithms for Mean Estimation under Local Differential Privacy

Hilal Asi^{*1} Vitaly Feldman^{*2} Kunal Talwar^{*2}

Theorem 2. *Let $\mathcal{R} : \mathbb{S}^{d-1} \rightarrow \mathcal{Z}$ be an ε -DP local randomizer, and $\mathcal{A} : \mathcal{Z}^n \rightarrow \mathbb{R}^d$ be an aggregation procedure such that $\mathbb{E}[\mathcal{A}(\mathcal{R}(v_1), \dots, \mathcal{R}(v_n))] = \frac{1}{n} \sum_{i=1}^n v_i$ for all $v_1, \dots, v_n \in \mathbb{S}^{d-1}$. Then there is $p^* \in [0, 1]$ and $\gamma^* \in [0, 1]$ such that $\text{PrivUnit}(p^*, \gamma^*)$ is ε -DP local randomizer and*

$$\text{Err}(\mathcal{A}^+, \text{PrivUnit}(p^*, \gamma^*)) \leq \text{Err}(\mathcal{A}, \mathcal{R}).$$

Privacy-Communication-Utility

Toward Exact Optimality

Order Optimal Results

$$O\left(\frac{d}{\min(\varepsilon^2, \varepsilon, b)}\right)$$

NeurIPS 2020

**Breaking the Communication-Privacy-Accuracy
Trilemma**

AISTATS 2022

Optimal Compression of Locally Differentially Private Mechanisms

ICML 2022

Algorithms for the Communication of Samples

Problem Setup

Each user has: v_i

Allow shared randomness: $U_1, \dots, U_n$

Encoder: $m_i = f_i(v_i, U_i)$

Server: $\mathcal{A}(m_1, \dots, m_n, U_1, \dots, U_n)$

Goal:

- **Rate: Minimum Bits**
- **Utility: Minimum MSE**
- **Privacy: LDP**

LDP with Shared Randomness

Local Differential Privacy (LDP) A randomized algorithm $\mathcal{Q} : \mathcal{X} \rightarrow \mathcal{Y}$ is ε -LDP if

$$\forall x, x' \in \mathcal{X}, y \in \mathcal{Y}, \frac{\mathcal{Q}(y|x)}{\mathcal{Q}(y|x')} \leq e^\varepsilon.$$

LDP with Shared Randomness In this work, we assume that the encoder and the decoder have access to a shared source of randomness $U \in \mathcal{U}$, where the random encoder (randomizer) $\mathcal{Q}$ privatizes x with additional randomness U . Then, the corresponding ε -LDP constraint is

$$\forall x, x' \in \mathcal{X}, y \in \mathcal{Y}, \frac{\mathcal{Q}(y|x, u)}{\mathcal{Q}(y|x', u)} \leq e^\varepsilon \tag{2}$$

for P_U -almost all u .

Mean Estimation: Privacy-Utility & Compression

DP-Encoder

$$f_i : \mathbb{S}^{d-1} \times \mathbb{R}^t \rightarrow [M], \text{ where } M = 2^b$$

Unbiased

Compression

$$\mathbb{E} [\mathcal{A}(m_1, \dots, m_n, U_1, \dots, U_n)] = \frac{1}{n} \sum_{i=1}^n v_i.$$

Utility

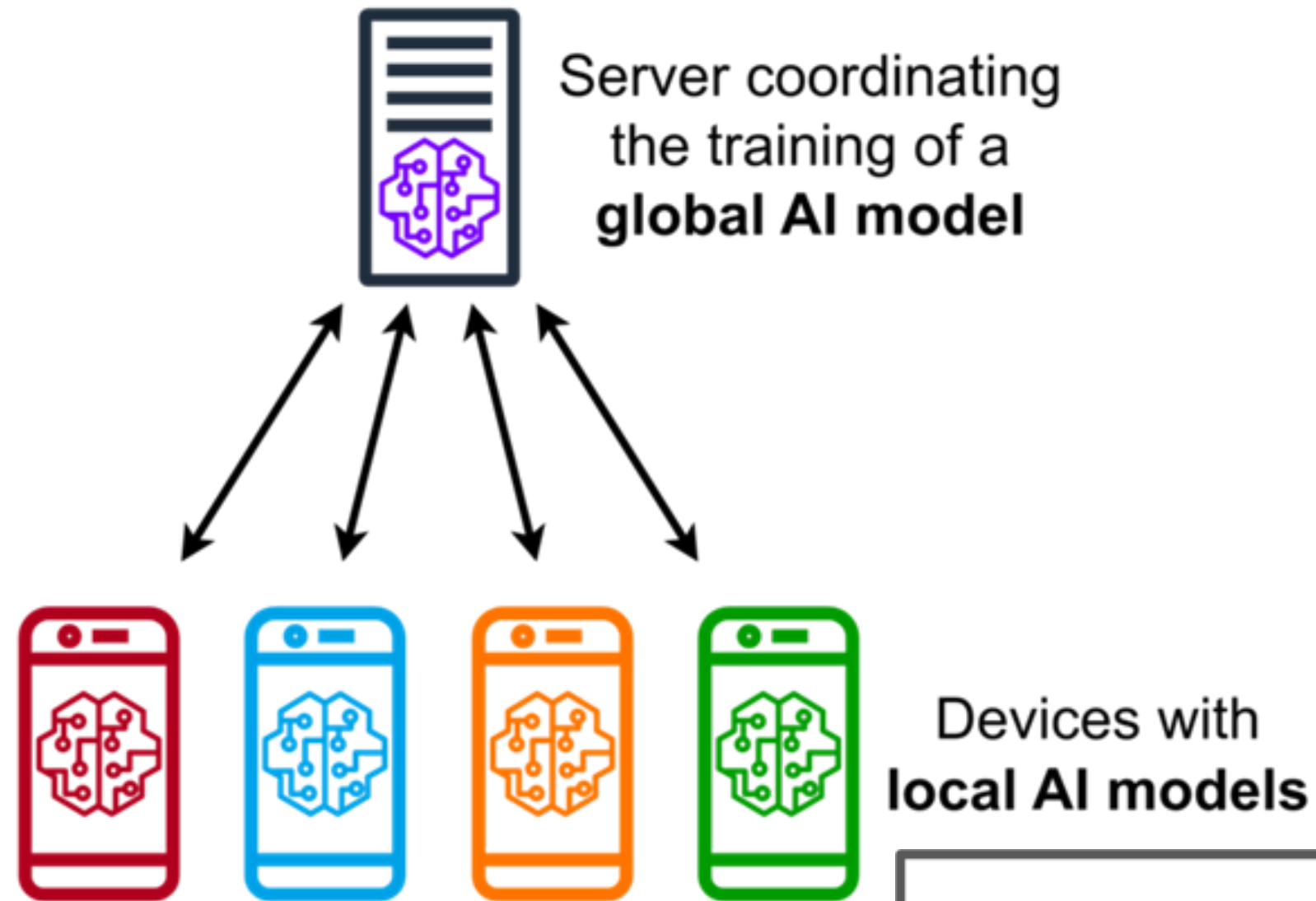
$$\text{Err}_n(f, \mathcal{A}, P_{U^n}) = \sup_{v_1, \dots, v_n \in \mathbb{S}^{d-1}} \mathbb{E} \left[\left\| \mathcal{A}(m_1, \dots, m_n, U_1, \dots, U_n) - \frac{1}{n} \sum_{i=1}^n v_i \right\|_2^2 \right]$$

Proof is mostly from

Optimal Algorithms for Mean Estimation under Local Differential Privacy

Hilal Asi^{*1} Vitaly Feldman^{*2} Kunal Talwar^{*2}

Main Result I: Canonical Protocols



$$m_i = f_i(v_i, U_i)$$

$$\mathcal{A}^+(m_1, \dots, m_n, U_1, \dots, U_n) = \frac{1}{n} \sum_{i=1}^n g_i(m_i, U_i).$$

Can decode separately

- **can focus single user case**

Main Result I: Canonical Protocols

Single Encoder

$$m = f(v, U)$$

Single Decoder

$$\hat{v} = g(m, U)$$

Error

$$\sup_v \mathbb{E}[\|v - \hat{v}\|^2]$$

Main Result II: Codebook based scheme

Optimal way of using shared randomness

Lemma 3.2. *For any f, g, P_U with $U \in \mathbb{R}^t$ that are unbiased and that satisfy ε -LDP, there exists a shared randomness $\tilde{U}^M \in (\mathbb{R}^d)^M$ and random encoder $f_0 : \mathbb{S}^{d-1} \times (\mathbb{R}^d)^M \rightarrow [M]$ such that*

$$D(v, f, g, P_U) = D(v, f_0, g^+, P_{\tilde{U}^M}) \quad (17)$$

for all $v \in \mathbb{S}^{d-1}$, where $f_0, g^+, P_{\tilde{U}^M}$ also satisfy unbiasedness and ε -LDP.

Codebook $U^M = (U_1, \dots, U_M) \in (\mathbb{R}^d)^M$

decoder $g^+(m, U^M) = U_m.$

Main Result II: Codebook based scheme

Optimal way of using shared randomness

**Codebook based
(random) encoder**

$$m = f(v, U^M)$$

**Codebook based
decoder**

$$\hat{v} = g(m, U^M) = U_m$$

Codebook

$$U_1, U_2, \dots, U_M \in \mathbb{R}^d$$

Main Result II: Codebook based scheme

Motivated from Coding Theory

**Codebook based
(random) encoder**

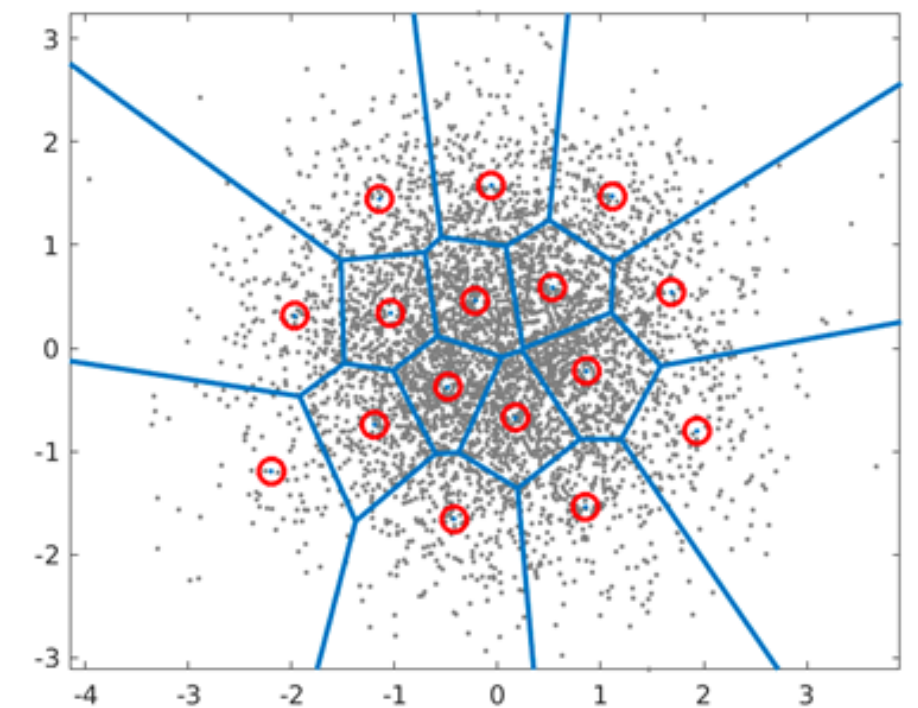
$$m = f(v, U^M)$$

**Codebook based
decoder**

$$\hat{v} = g(m, U^M) = U_m$$

Codebook

$$U_1, U_2, \dots, U_M \in \mathbb{R}^d$$



Main Result II: Rotationally Symmetric Codebook

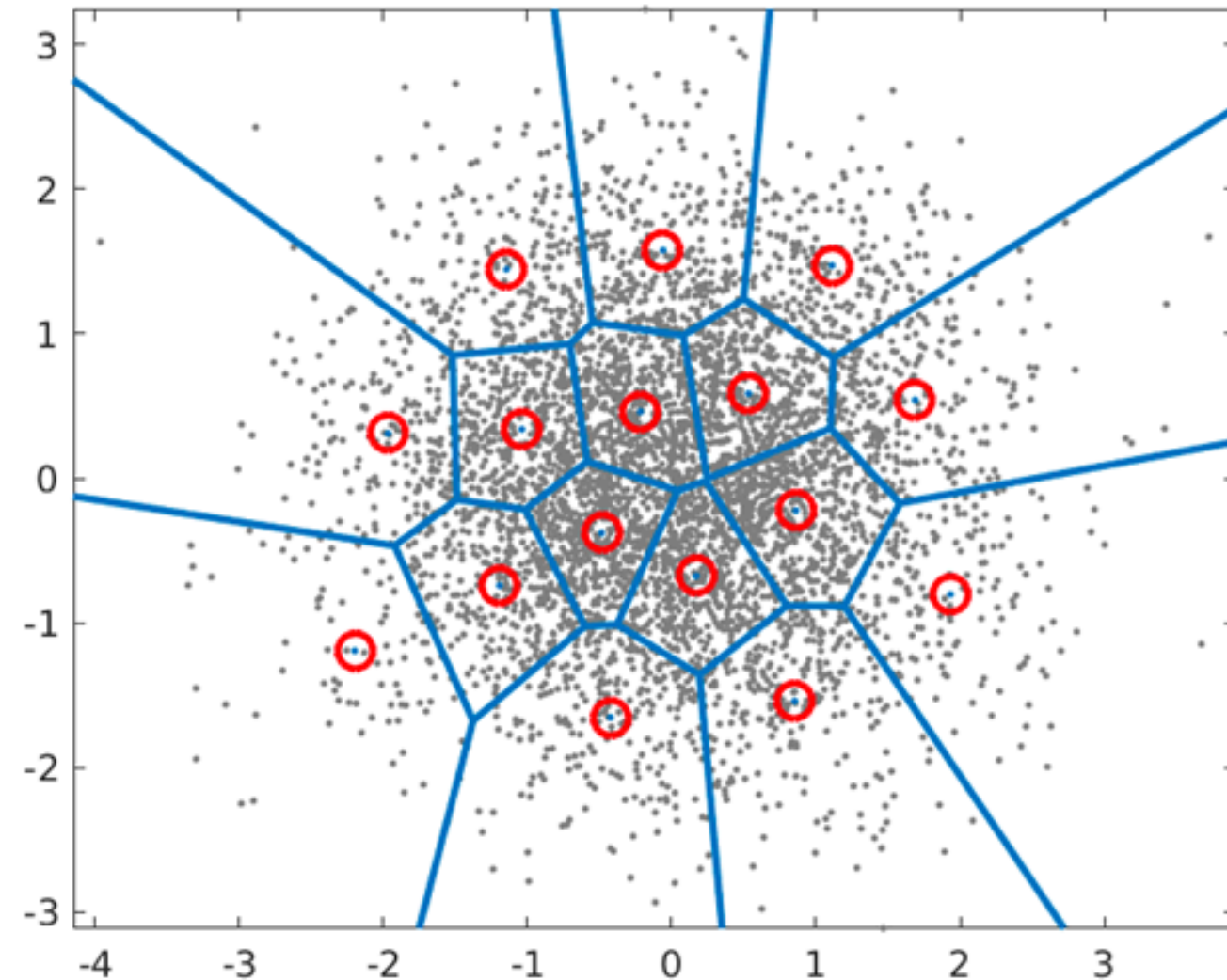
Lemma 3.4. *Let P_{U^M} be a codebook generating distribution, and suppose random encoder f satisfies unbiasedness and ε -LDP. Then, there exists a random encoder f_1 and rotationally symmetric random codebook $\bar{U}^M$ such that*

$$\text{Err}(f, P_{U^M}) \geq \text{Err}(f_1, P_{\bar{U}^M}), \quad (21)$$

which also satisfies unbiasedness and ε -LDP.

Definition 3.3. A random codebook $U^M \in (\mathbb{R}^d)^M$ is **rotationally symmetric** if $(U_1, \dots, U_M) \stackrel{(d)}{=} (A_0 U_1, \dots, A_0 U_M)$ for any $d \times d$ orthonormal matrix A_0 .

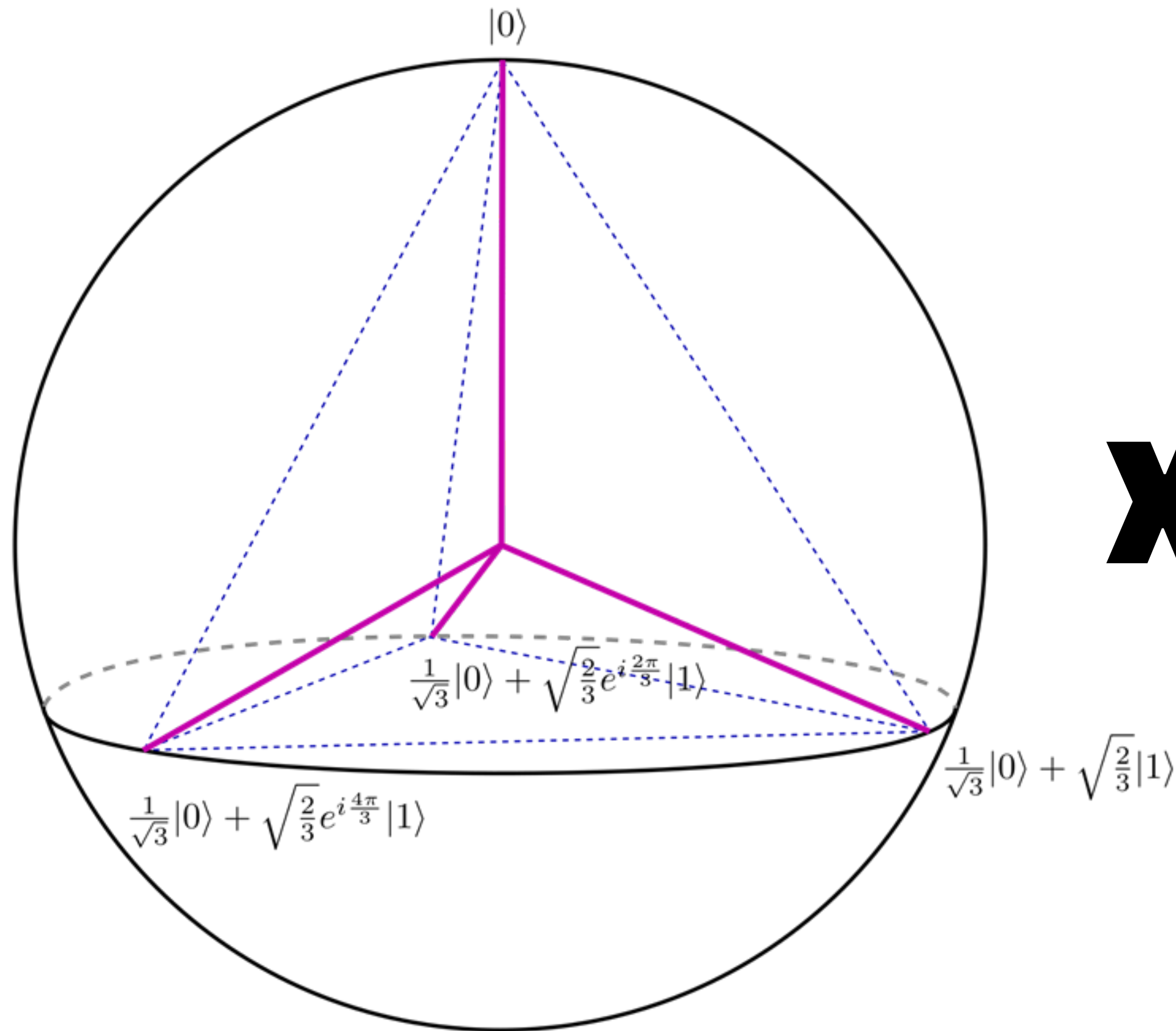
Codebook Design is usually Challenging



**In Compression problems,
 $M \gg d$ and it is hard**

What if $M < d$?

Rotationally Symmetric Symplex Codebook (RRSC)



X

**(random orthogonal
matrix)**

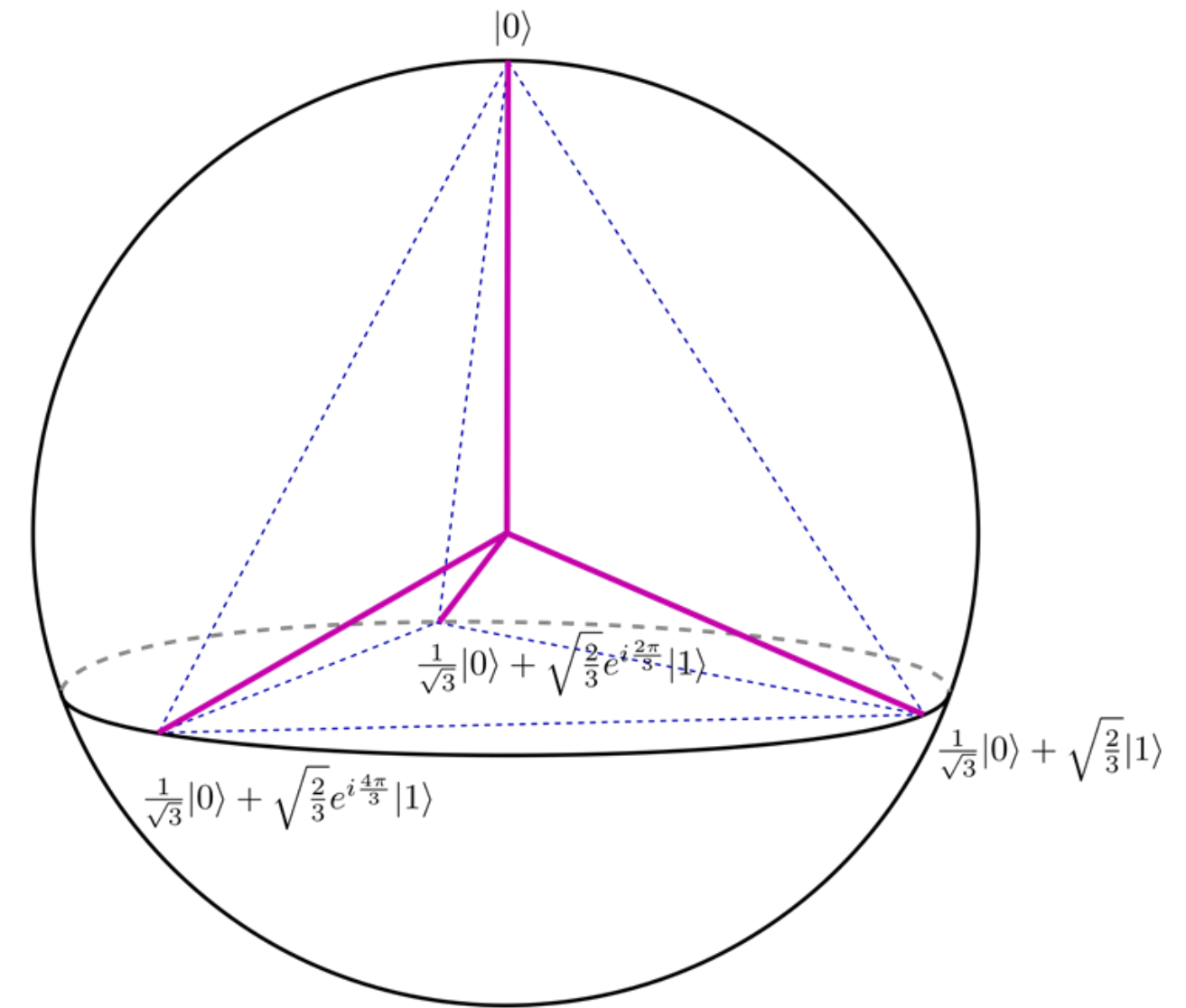
Rotationally Symmetric Symplex Codebook (RRSC)

$$(s_i)_j = \begin{cases} \frac{M-1}{\sqrt{M(M-1)}} & \text{if } i = j \\ -\frac{1}{\sqrt{M(M-1)}} & \text{if } i \neq j \text{ and } j \leq M \\ 0 & \text{if } j > M \end{cases}$$

$$(U_1, U_2, \dots, U_M) \stackrel{(d)}{=} (rAs_1, rAs_2, \dots, rAs_M).$$

Main Result IV: k-closest encoding

```
for  $m \in [M]$  do  
  if  $\langle v, U_m \rangle$  is one of the  $k$  largest then  
     $Q_f(m|v, U^M) \leftarrow \frac{e^\varepsilon}{ke^\varepsilon + (M-k)}$   
  else  
     $Q_f(m|v, U^M) \leftarrow \frac{1}{ke^\varepsilon + (M-k)}$   
  end if  
end for
```



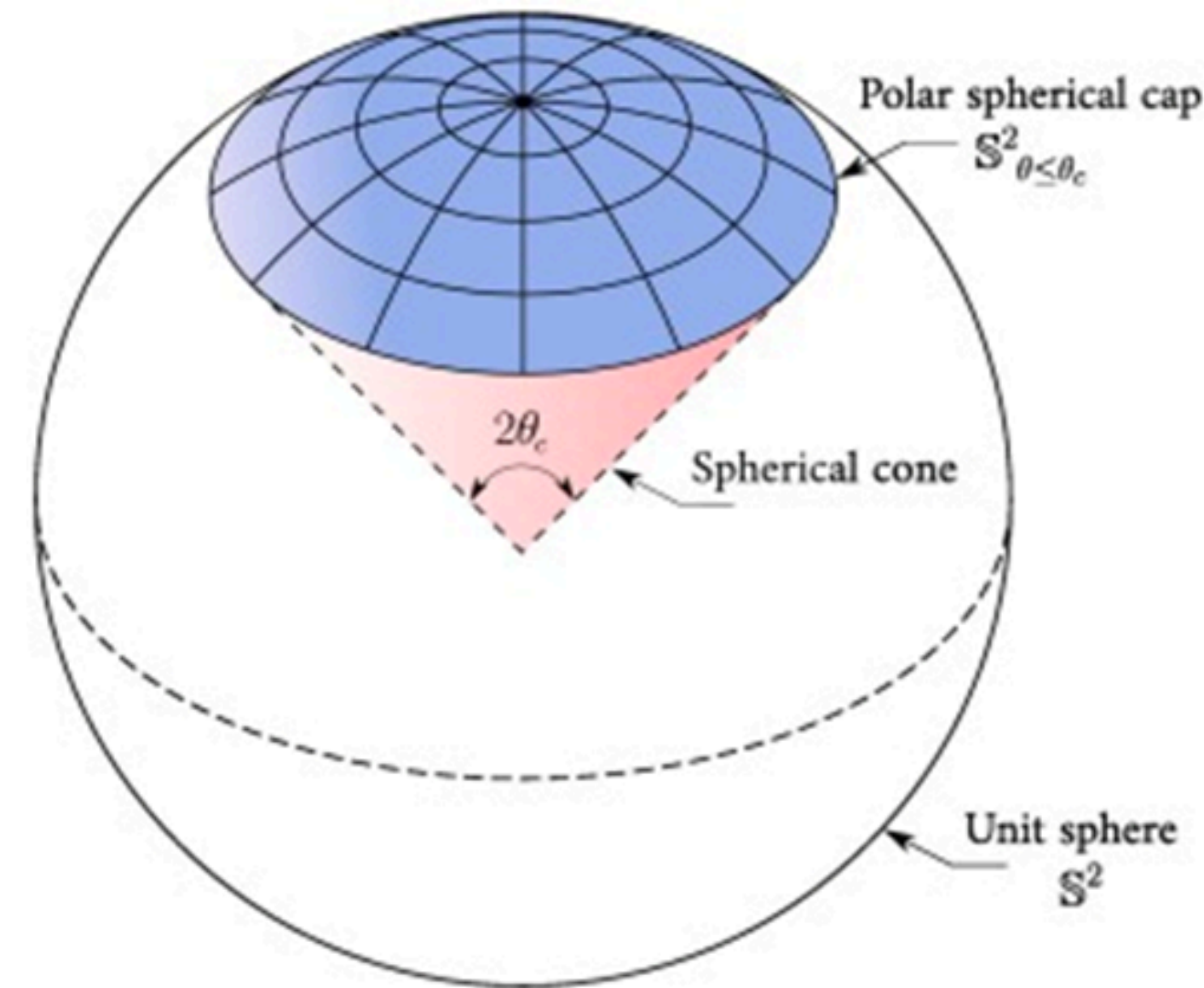
It is optimal for RRSC

RRSC to PrivUnit

What if M increases

```
for  $m \in [M]$  do  
  if  $\langle v, U_m \rangle$  is one of the  $k$  largest then  
     $Q_f(m|v, U^M) \leftarrow \frac{e^\epsilon}{ke^\epsilon + (M-k)}$   
  else  
     $Q_f(m|v, U^M) \leftarrow \frac{1}{ke^\epsilon + (M-k)}$   
  end if  
end for
```

RRSC



PrivUnit

Unified Framework

$$O\left(\frac{d}{\min(\varepsilon^2, \varepsilon, b)}\right)$$

NeurIPS 2020

**Breaking the Communication-Privacy-Accuracy
Trilemma**

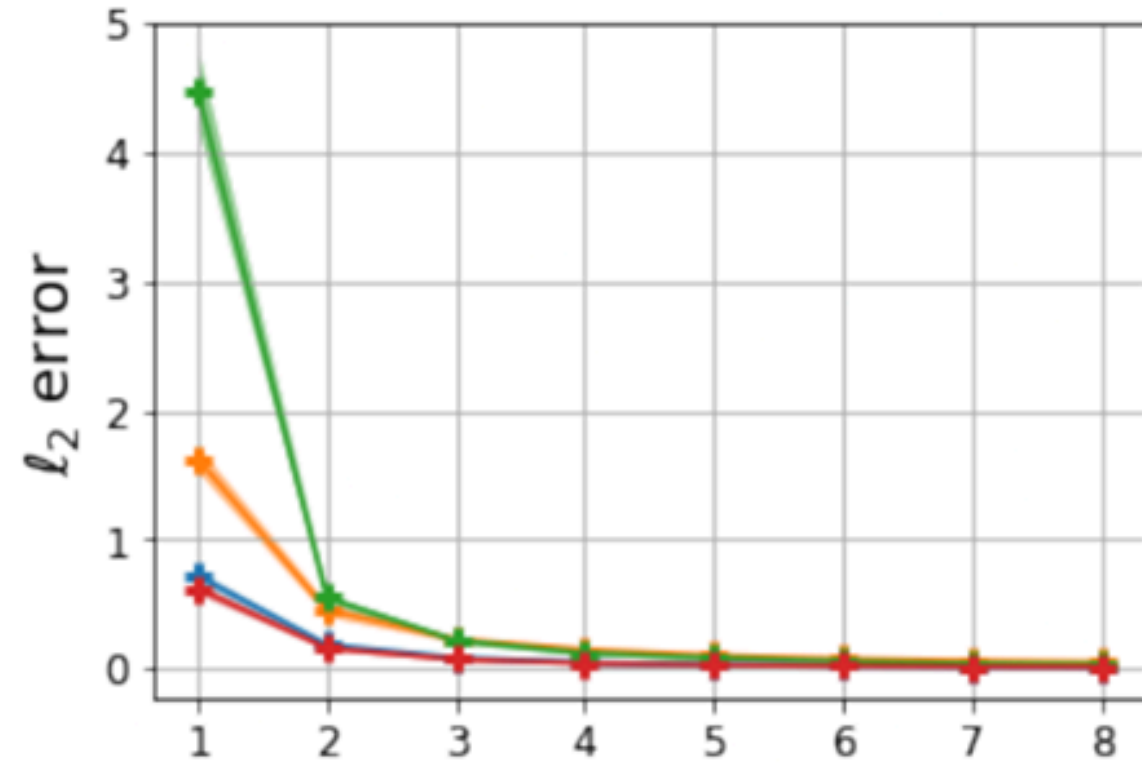
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Optimal Compression of Locally Differentially Private Mechanisms

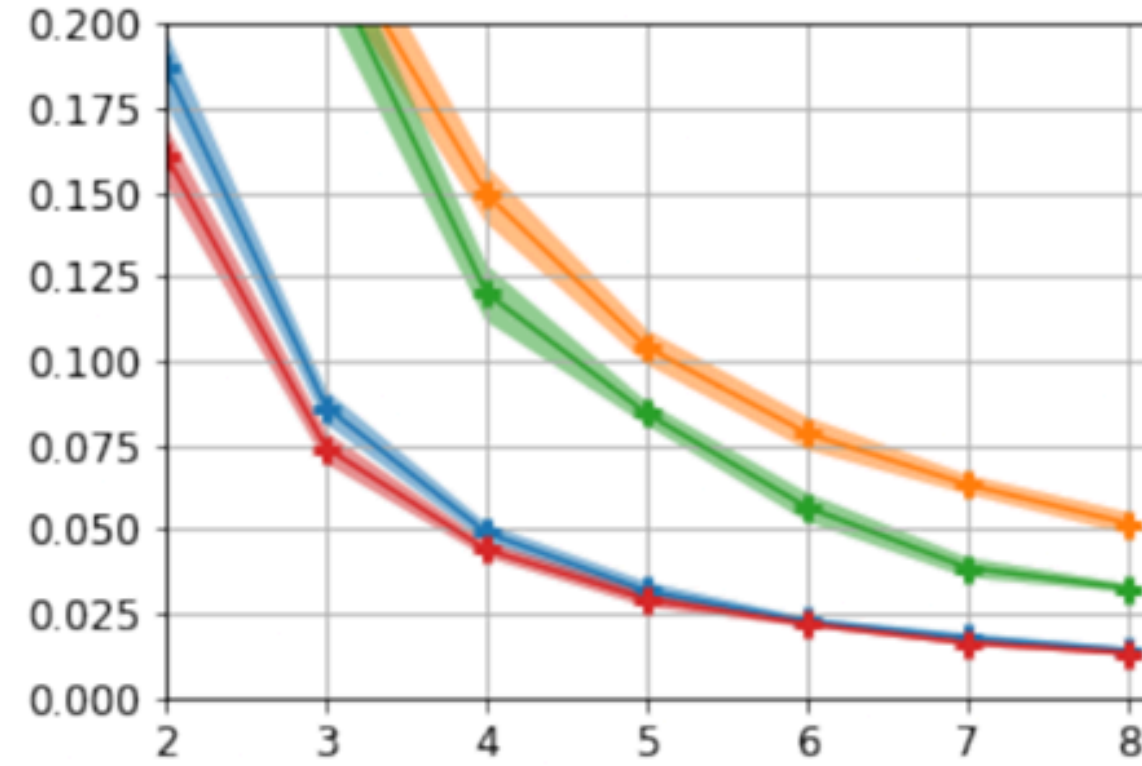
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Algorithms for the Communication of Samples

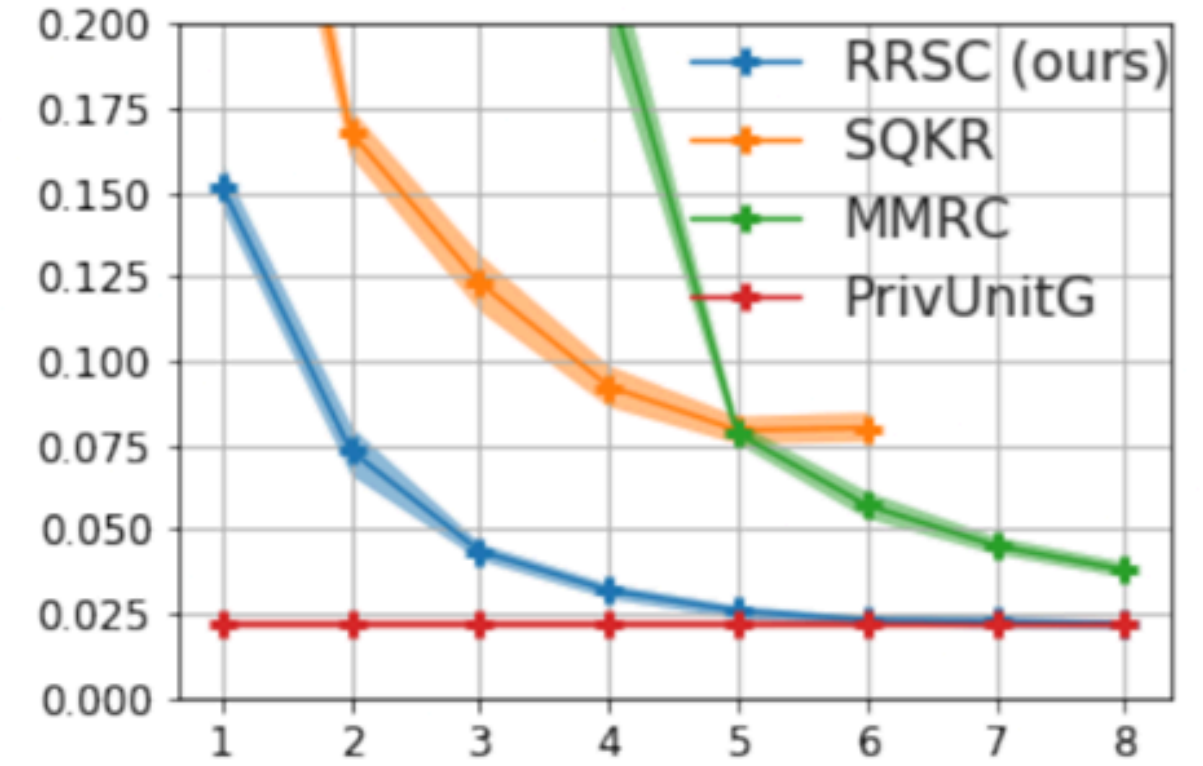
Experiments



$b = 6$



$b = 6$



$\epsilon = 6$

$n = 5000$ users

$d = 500$ dimensional vector

Conjecture

Conjecture 5.1. *The proposed k -closest encoding is exact-optimal for any rotationally symmetric codebook.*

Conjecture 5.2. *The rotationally symmetric simplex codebook is the exact-optimal codebook.*

Questions?

Alternative DP

Definition 3 (Rényi divergence). For two probability distributions P and Q defined over $\mathcal{R}$, the Rényi divergence of order $\alpha > 1$ is

$$D_{\alpha}(P\|Q) \triangleq \frac{1}{\alpha - 1} \log \mathbb{E}_{x \sim Q} \left(\frac{P(x)}{Q(x)} \right)^{\alpha}.$$

Definition 4 $((\alpha, \epsilon)$ -RDP). A randomized mechanism $f: \mathcal{D} \mapsto \mathcal{R}$ is said to have ϵ -Rényi differential privacy of order α , or (α, ϵ) -RDP for short, if for any adjacent $D, D' \in \mathcal{D}$ it holds that

$$D_{\alpha}(f(D)\|f(D')) \leq \epsilon.$$

Renyi Differential Privacy

Diffusion + DP

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Differentially Private Diffusion Models

Tim Dockhorn

*Stability AI **

timdockhorn@gmail.com

Tianshi Cao

NVIDIA

University of Toronto

Vector Institute

tianshic@nvidia.com

Arash Vahdat

NVIDIA

avahdat@nvidia.com

Karsten Kreis

NVIDIA

kkreis@nvidia.com

Reviewed on OpenReview: <https://openreview.net/forum?id=ZPpQk7FJXF>



Figure 5: Fashion-MNIST images generated by, from top to bottom, DP-CGAN (Torkzadehmahani et al., 2019), DP-MERF (Harder et al., 2022), Datalens (Wang et al., 2021), G-PATE (Long et al., 2019), GS-WGAN (Chen et al., 2020), DP-Sinkhorn (Cao et al., 2021), PEARL (Liew et al., 2022), DPGANr (Bie et al., 2022) (all above bar), and our DPDM (below bar) using the privacy budget $\epsilon=10$. See App. F.5 for more samples.

DP-RDM: Adapting Diffusion Models to Private Domains Without Fine-Tuning

Jonathan Lebensold^{1,2,3}, Maziar Sanjabi¹, Pietro Astolfi¹, Adriana Romero-Soriano^{1,2,3,4}, Kamalika Chaudhuri¹, Mike Rabbat^{1,2}, Chuan Guo¹

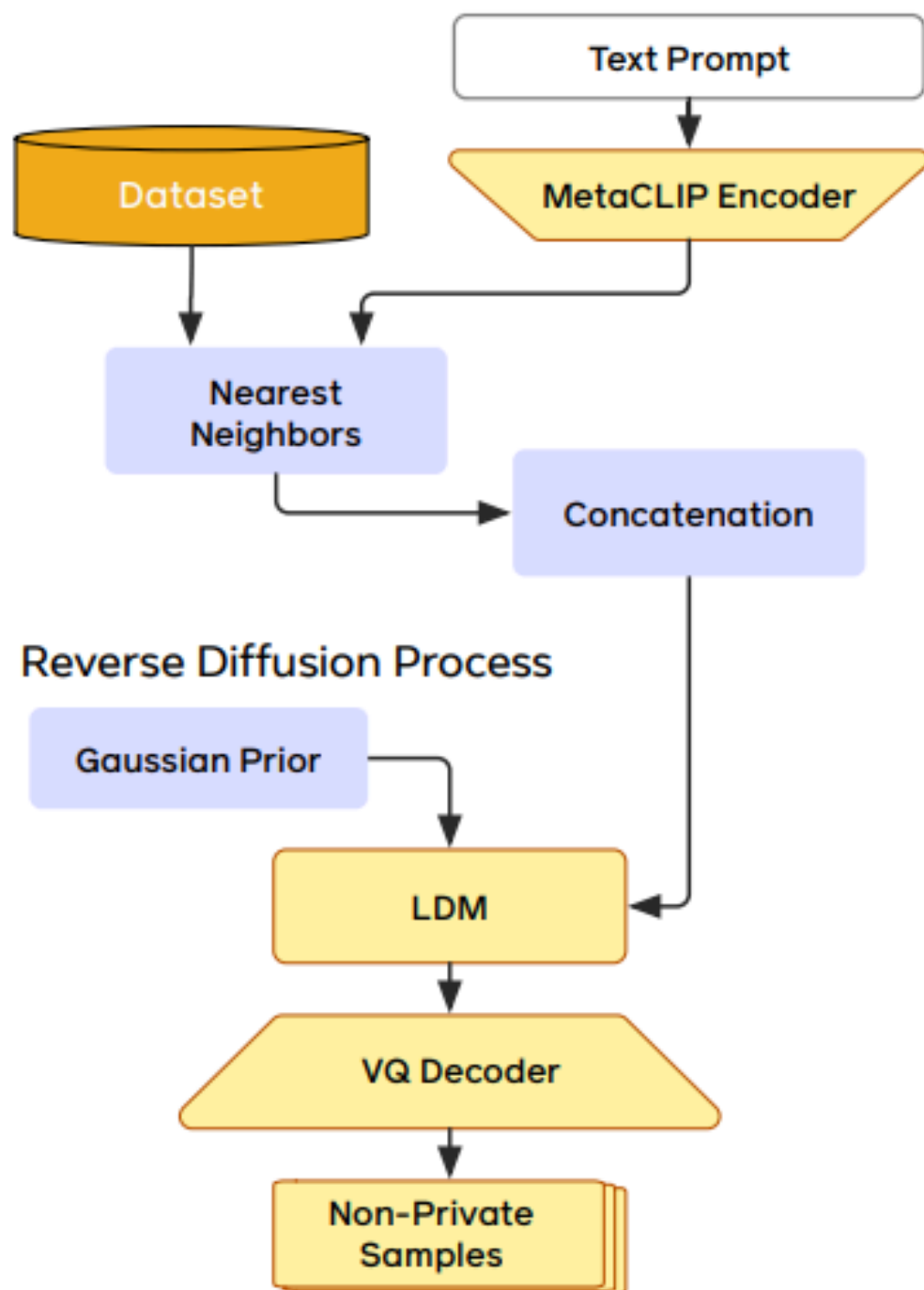
¹Meta AI, ²Mila - Québec AI Institute, Montreal, Quebec, Canada, ³Department of Computer Science, McGill University, Montreal, Quebec, Canada, ⁴Canada CIFAR AI Chair

Text-to-image diffusion models have been shown to suffer from sample-level memorization, possibly reproducing near-perfect replica of images that they are trained on, which may be undesirable. To remedy this issue, we develop the first differentially private (DP) retrieval-augmented generation algorithm that is capable of generating high-quality image samples while providing provable privacy guarantees. Specifically, we assume access to a text-to-image diffusion model trained on a small amount of public data, and design a DP retrieval mechanism to augment the text prompt with samples retrieved from a private retrieval dataset. Our *differentially private retrieval-augmented diffusion model* (DP-RDM) requires no fine-tuning on the retrieval dataset to adapt to another domain, and can use state-of-the-art generative models to generate high-quality image samples while satisfying rigorous DP guarantees. For instance, when evaluated on MS-COCO, our DP-RDM can generate samples with a privacy budget of $\epsilon = 10$, while providing a 3.5 point improvement in FID compared to public-only retrieval for up to 10,000 queries.

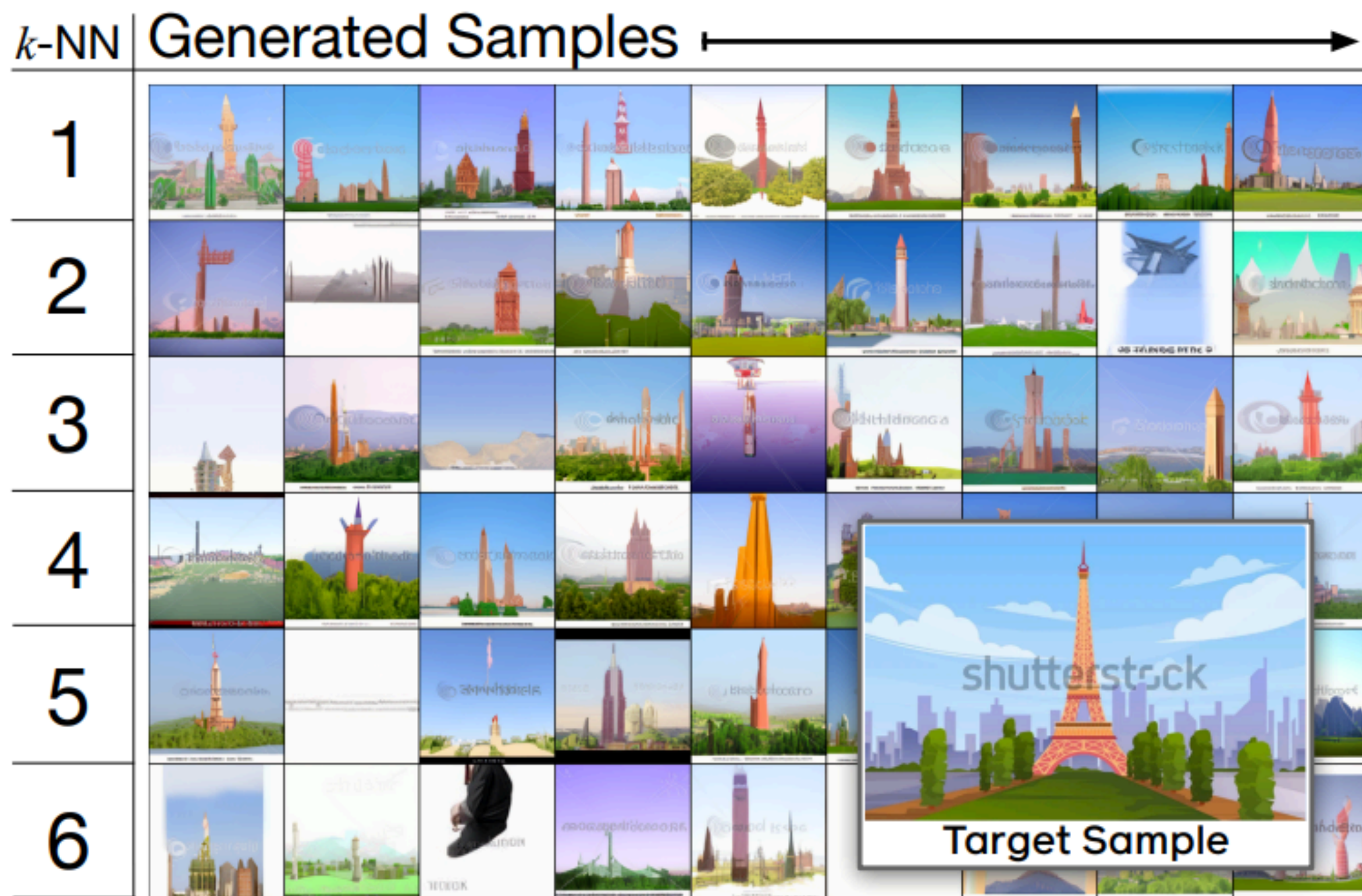
Date: April 17, 2024

Correspondence: Jonathan Lebensold at lebensold@meta.com, Chuan Guo at chuanguo@meta.com



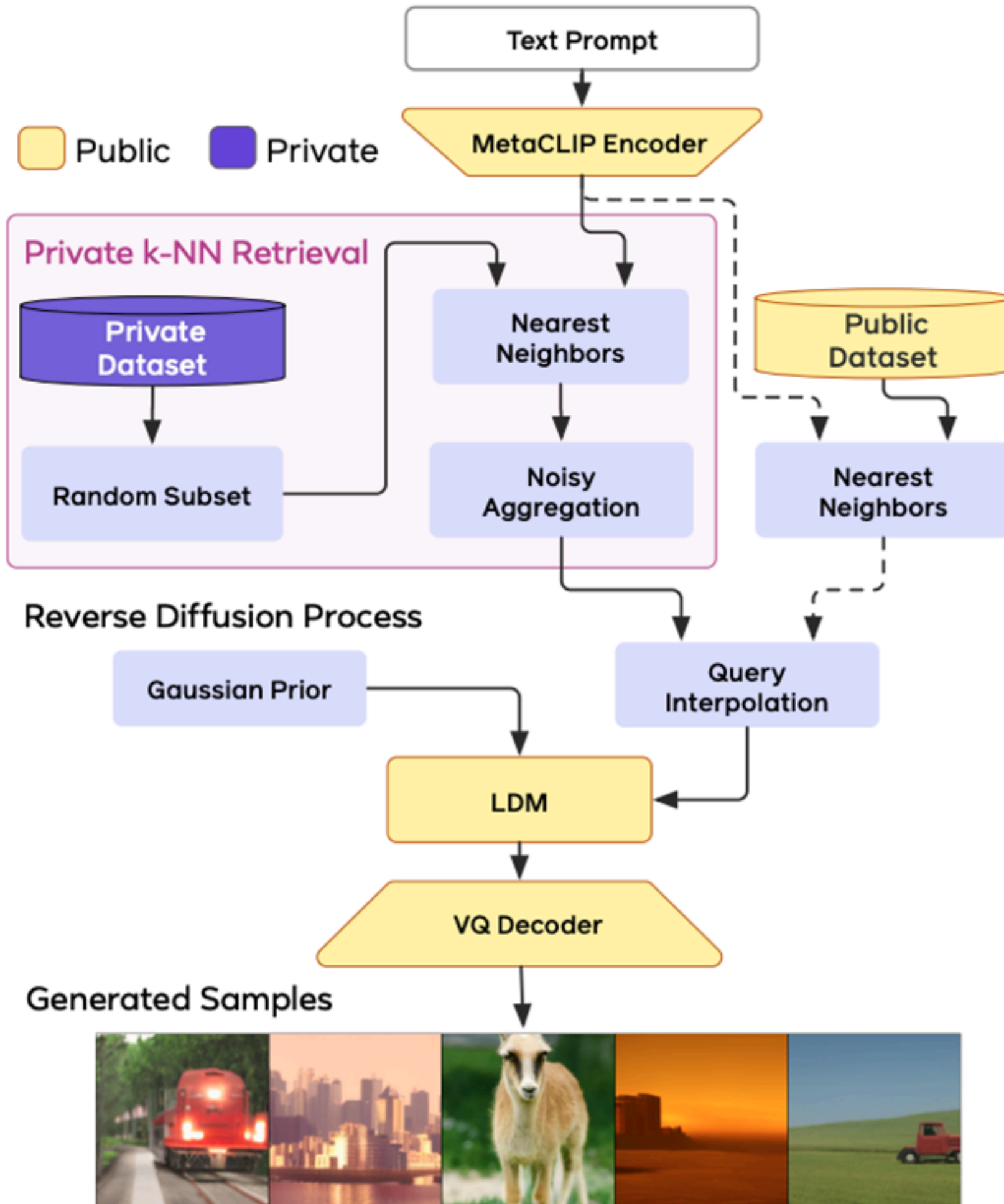


(a) RDM architecture.



(b) Samples generated by RDM using an adversarial retrieval dataset.

Figure 2 (a) RDM architecture from Blattmann et al. (2022). (b) Samples generated with a non-private RDM. The retrieval dataset consists of blank images and one illustration of the Eiffel Tower with a Shutterstock watermark. Each row shows samples for a different number of retrieved neighbors k . The watermark is clearly visible even though it came only from conditioning on the retrieval dataset.



Algorithm 1 Generating Samples with DP-RDM.

input: LDM ϕ , public retrieval dataset U ; private retrieval dataset D ; text prompt $\mathbf{y} \in \mathbb{R}^d$; prior $\xi \sim \mathcal{N}(0, I)$

input: Noise parameter $\sigma > 0$, num. of neighbors $k \in \mathbb{N}$, query interpolation $\lambda \in [0, 1]$, subsample rate $q \in [0, 1]$.

k -NN retrieval:

$D' = \text{RandomSubset}(D, q)$

$N_k = \text{NearestNeighbors}(D', \mathbf{y}, k)$

$O_k = \text{NearestNeighbors}(U, \mathbf{y}, k)$

Noisy aggregation:

$\mathbf{z} = \frac{1}{k} \left(\sum_{j=1}^k N_k^{(j)} \right) + \mathcal{N}(0, \sigma^2 I)$

Query interpolation:

$\mathbf{e} = (1 - \lambda) \cdot O_k + \lambda \cdot [\mathbf{z} \mid \cdots \mid \mathbf{z}]$

return $\phi(\mathbf{e}, \xi)$
